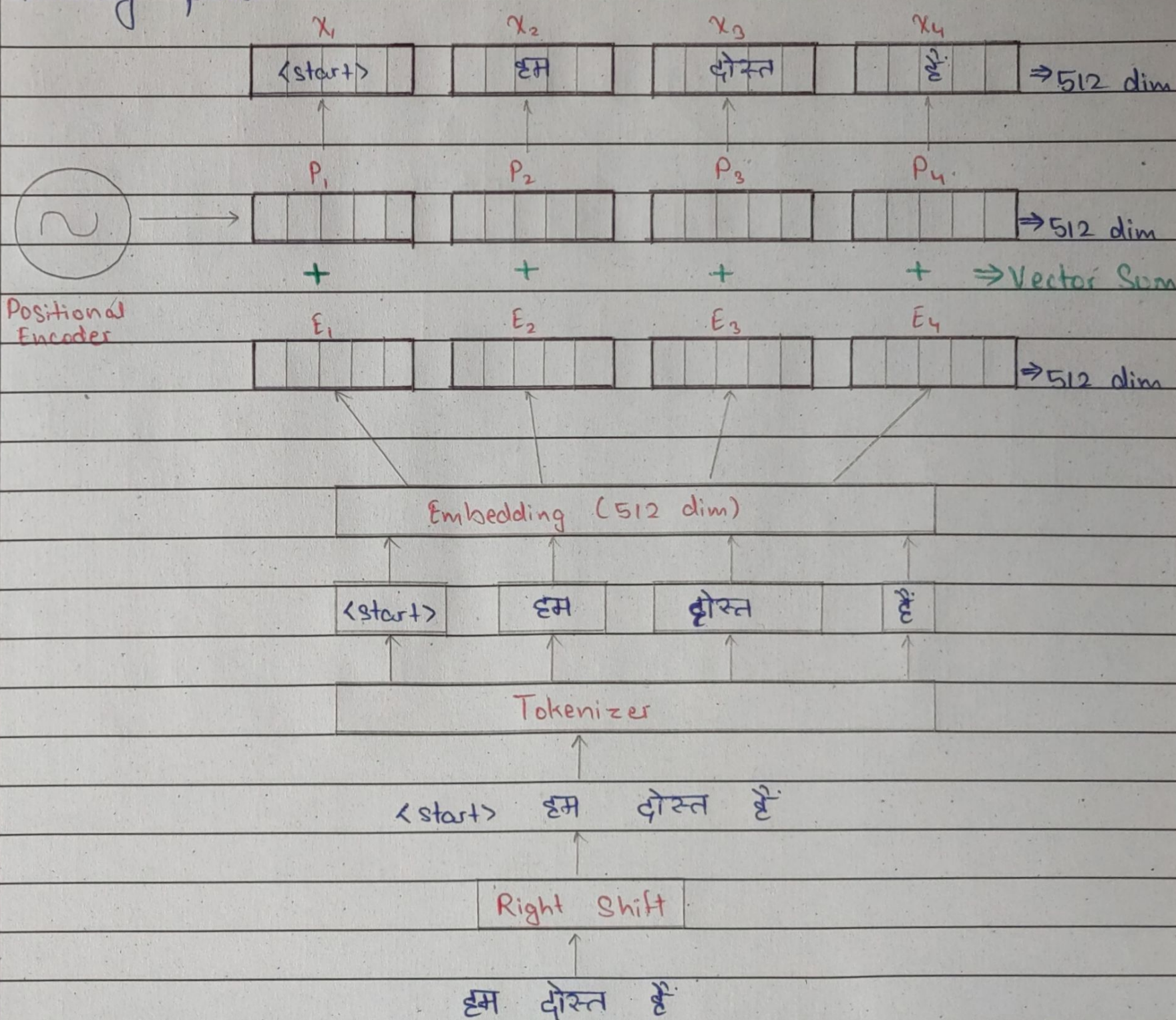




Consider the task of English to Hindi Machine Translation. The English sentence "We are friends" is to be translated to its Hindi form "हम दोस्त हैं". So the English sentence is passed through the Encoder which generates its contextual or encoded embeddings as outputs & pass it to the decoder. The Hindi sentence in which the English sentence is to be translated is passed as input to the decoder during the training phase.





- * The Hindi text is first shifted right, i.e. an additional $\langle \text{start} \rangle$ token is appended at the beginning of the text.
- * The $\langle \text{start} \rangle$ token tells the decoder to start decoding the encoded sequence received from the encoder.
- * The resultant text is then tokenized by the tokenizer to feed into the Embedding layer.
- * The Embedding layer converts the tokenized words into 512 dimensional embedding vectors $\Rightarrow E_1, E_2, E_3, E_4$.
- * A 512 dimensional positional encoding is calculated of each word & added to the embedding vectors $\Rightarrow P_1 + E_1, P_2 + E_2, P_3 + E_3, P_4 + E_4$.
- * The resultant position encoded embedding vectors are fed into the $\&$ Decoder block $\Rightarrow x_1, x_2, x_3, x_4$.